

TESTING REQUIREMENTS

For Challenge 1, test the following:

1. At least one case with a positive slope:

```
/usr/bin/python3 /Users/francisobille/Documents/APPDAET-CC2/Challenge01_TwoPointLineAnalyzer.py
francisobille@nadbook APPDAET-CC2 % /usr/bin/python3 /Users/francisobille/Documents/APPDAET-CC2/Challenge01_TwoPoi
ntLineAnalyzer.py
=====
CHALLENGE 1: TWO-POINT LINE ANALYZER
=====
Enter Point 1 coordinates:
Enter x1: 2
Enter y1: 3

Enter Point 2 coordinates:
Enter x2: 6
Enter y2: 11

=====
ANALYSIS RESULTS
=====
Change in x ( $\Delta x$ ) : 4.0
Change in y ( $\Delta y$ ) : 8.0
Slope (m) : 2.0
X-intercept : (0.5, 0)
Y-intercept : (0, -1.0)
=====
Slope-intercept form :  $y = 2.0x + -1.0$ 
Point-slope form :  $y - 3.0 = 2.0(x - 2.0)$ 
General form :  $2.0x + (-1.0)y + (-1.0) = 0$ 
=====
Midpoint : (4.0, 7.0)
Distance : 8.94427190999916
=====
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```

2. One case with a negative slope:

```
/usr/bin/python3 /Users/francisobille/Documents/APPDAET-CC2/Challenge01_TwoPointLineAnalyzer.py
francisobille@nadbook APPDAET-CC2 % /usr/bin/python3 /Users/francisobille/Documents/APPDAET-CC2/Challenge01_TwoPoi
ntLineAnalyzer.py
=====
CHALLENGE 1: TWO-POINT LINE ANALYZER
=====
Enter Point 1 coordinates:
Enter x1: 1
Enter y1: 8

Enter Point 2 coordinates:
Enter x2: 5
Enter y2: 4

=====
ANALYSIS RESULTS
=====
Change in x ( $\Delta x$ ) : 4.0
Change in y ( $\Delta y$ ) : -4.0
Slope (m) : -1.0
X-intercept : (9.0, 0)
Y-intercept : (0, 9.0)
=====
Slope-intercept form :  $y = -1.0x + 9.0$ 
Point-slope form :  $y - 8.0 = -1.0(x - 1.0)$ 
General form :  $-1.0x + (-1.0)y + (9.0) = 0$ 
=====
Midpoint : (3.0, 6.0)
Distance : 5.656854249492381
=====
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```

3. One case containing decimal coordinates.

```
/usr/bin/python3 /Users/francisobille/Documents/APPDAET-CC2/Challenge01_TwoPointLineAnalyzer.py
francisobille@nadbook APPDAET-CC2 % /usr/bin/python3 /Users/francisobille/Documents/APPDAET-CC2/Challenge01_TwoPointLineAnalyzer.py

=====
CHALLENGE 1: TWO-POINT LINE ANALYZER
=====

Enter Point 1 coordinates:
Enter x1: 1.5
Enter y1: 2.5

Enter Point 2 coordinates:
Enter x2: 4.5
Enter y2: 8.5

=====
ANALYSIS RESULTS
=====
Change in x ( $\Delta x$ ) : 3.0
Change in y ( $\Delta y$ ) : 6.0
Slope (m) : 2.0
X-intercept : (0.25, 0)
Y-intercept : (0, -0.5)
=====
Slope-intercept form :  $y = 2.0x + -0.5$ 
Point-slope form :  $y - 2.5 = 2.0(x - 1.5)$ 
General form :  $2.0x + (-1.0)y + (-0.5) = 0$ 
=====
Midpoint : (3.0, 5.5)
Distance : 6.708203932499369
=====
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```

For Challenge 2, test the following:

1. Test the instructor-provided matrix [verify determinant = 1 and trace = 2]:

```
/usr/bin/python3 /Users/francisobille/Documents/APPDAET-CC2/Challenge02_MatrixAnalyzer.py
francisobille@nadbook APPDAET-CC2 % /usr/bin/python3 /Users/francisobille/Documents/APPDAET-CC2/Challenge02_MatrixAnalyzer.py

=====
3 x 3 MATRIX ANALYZER
=====

Enter a11: 1
Enter a12: 0
Enter a13: 5
Enter a21: 2
Enter a22: 1
Enter a23: 6
Enter a31: 3
Enter a32: 4
Enter a33: 0

ORIGINAL MATRIX
| 1.00 0.00 5.00 |
| 2.00 1.00 6.00 |
| 3.00 4.00 0.00 |

MATRIX PROPERTIES
Determinant: 1.00
Trace: 2.00

TRANSPOSE
| 1.00 2.00 3.00 |
| 0.00 1.00 4.00 |
| 5.00 6.00 0.00 |
=====
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```

2. One matrix created by your group:

```
/usr/bin/python3 /Users/francisobille/Documents/APPDAET-CC2/Challenge02_MatrixAnalyzer.py
francisobille@nadbook APPDAET-CC2 % /usr/bin/python3 /Users/francisobille/Documents/APPDAET-CC2/Challenge02_MatrixAnalyzer.py

=====
          3 x 3 MATRIX ANALYZER
=====

Enter a11: 3
Enter a12: 2
Enter a13: 7
Enter a21: 6
Enter a22: 0
Enter a23: 1
Enter a31: 0
Enter a32: 2
Enter a33: 3

ORIGINAL MATRIX
| 3.00 2.00 7.00 |
| 6.00 0.00 1.00 |
| 0.00 2.00 3.00 |

MATRIX PROPERTIES
Determinant: 42.00
Trace:      6.00

TRANSPOSE
| 3.00 6.00 0.00 |
| 2.00 0.00 2.00 |
| 7.00 1.00 3.00 |
=====

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```

3. One matrix containing decimal and/or negative values:

```
/usr/bin/python3 /Users/francisobille/Documents/APPDAET-CC2/Challenge02_MatrixAnalyzer.py
francisobille@nadbook APPDAET-CC2 % /usr/bin/python3 /Users/francisobille/Documents/APPDAET-CC2/Challenge02_MatrixAnalyzer.py

=====
          3 x 3 MATRIX ANALYZER
=====

Enter a11: -4
Enter a12: -9
Enter a13: -1
Enter a21: -2
Enter a22: -4
Enter a23: -3
Enter a31: -2
Enter a32: -1
Enter a33: -3

ORIGINAL MATRIX
| -4.00 -9.00 -1.00 |
| -2.00 -4.00 -3.00 |
| -2.00 -1.00 -3.00 |

MATRIX PROPERTIES
Determinant: -30.00
Trace:      -11.00

TRANSPOSE
| -4.00 -2.00 -2.00 |
| -9.00 -4.00 -1.00 |
| -1.00 -3.00 -3.00 |
=====

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```

```
/usr/bin/python3 /Users/francisobille/Documents/APPDAET-CC2/Challenge02_MatrixAnalyzer.py
francisobille@nadbook APPDAET-CC2 % /usr/bin/python3 /Users/francisobille/Documents/APPDAET-CC2/Challenge02_Matrix
Analyzer.py
=====
      3 x 3 MATRIX ANALYZER
=====

Enter a11: 1.2
Enter a12: 3.2
Enter a13: 4.1
Enter a21: 3.8
Enter a22: 1.1
Enter a23: 6.7
Enter a31: 2.3
Enter a32: 1.2
Enter a33: 5.9

ORIGINAL MATRIX
| 1.20 3.20 4.10 |
| 3.80 1.10 6.70 |
| 2.30 1.20 5.90 |

MATRIX PROPERTIES
Determinant: -15.97
Trace:      8.20

TRANSPOSE
| 1.20 3.80 2.30 |
| 3.20 1.10 1.20 |
| 4.10 6.70 5.90 |
=====
francisobille@nadbook APPDAET-CC2 %
```